Karnataka Veterinary, Animal and Fisheries Sciences University, Bidar

Department of Veterinary Microbiology

VETERINARY COLLEGE, HASSAN

A GUIDE TO DISEASE DIAGNOSIS



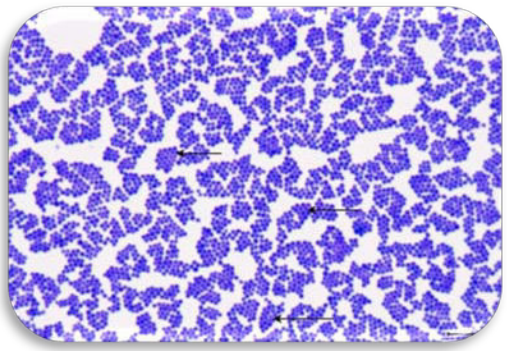
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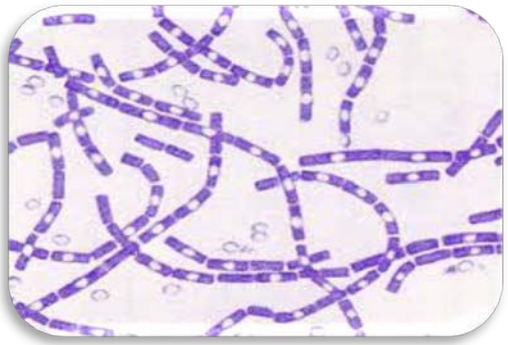
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Staphylococcus
(Cocci in clusters)



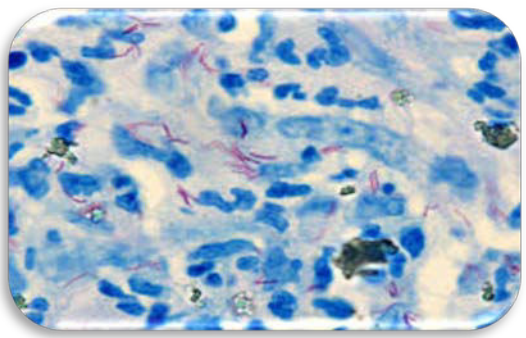
Streptococcus
(Cocci in long chains)



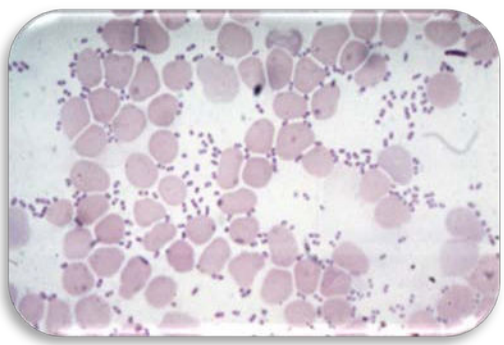
Bacillus anthracis
(Bacilli in chains)



Clostridium tetani
(Terminal spores – Tennis racket appearence)



Mycobacterium tuberculosis
(Acid Fast staining – Red colored organisms)



Pasteurella ssp.
(Bipolar organisms)

1. BACTERIAL DISEASES

Sl. No.	Disease / Condition	Causative agent	Clinical signs	Samples	Diagnostic tests (Acc. to OIE)	Treatment
1	Gangrenous Mastitis (Cattle), Botryomycosis (Horses), Bumble Foot (Poultry)	*Staphylococcus aureus* (G+ve Cocci)	Reduction in milk, Pyogenic lesions,	Milk, Pus, Exudate	Staining & Culture	Norfloxacin, Erythromycin, Ciprofloxacin, Vancomycin
2	Strangles (Horses)	*Streptococcus equi* ssp. *equi* (G+ve Cocci)	Catarrhal nasal discharges, swelling of pharyngeal lymph nodes	Pus	Staining & Culture	Procaine Penicillin
3	Anthrax	*Bacillus anthracis* (Large, G +ve aerobic bacilli)	Sudden death with oozing of blood from natural orifices, absence of rigor mortis, Carcass putrifies rapidly	Peripheral blood, Peritoneal fluid,	Mcfadyean`s reaction, FAT	Penicillin G, Oxytetracycline, Erythromycin
4	Black Quarter / Black Leg	*Clostridium chauvoei* (Large, G +ve anaerobic bacilli)	Lameness, patch of discoloration with crepitate swelling which crackles when rubbed on the affected muscles	Exudates, Muscle piece,	Staining & FAT	Penicillin, Tetracycline
5	Enterotoxemia (Sheep)	*Clostridium perfringens* Type D (G +ve anaerobic bacilli)	Dysentery bloat, death	Faeces, Intestinal contents, Serum	Staining & Culture	Penicillin, Chlortetracycline
6	Tetanus	*Clostridium tetani* (G +ve anaerobic bacilli)	Muscular spasms (Spastic paralysis), Lock jaw condition, body stiffness	Necrotic lesions, Serum	Demonstration of toxin, Mouse inoculation	Antitoxin, penicillins, Sedatives, Muscle relaxants
7	Botulism	*Clostridium botulinum* (G +ve anaerobic bacilli)	Flaccid paralysis of muscles, unable to swallow & protruding tongue, inability to hold neck in long necked birds	Contaminated food, Serum, GI contents	Demonstration of toxin, Mouse inoculation	Polyvalent antisera, Tetraethylamide & guanidine HCL

Sl. No.	Disease / Condition	Causative agent	Clinical signs	Samples	Diagnostic tests (Acc. to OIE)	Treatment
8	Circling Disease / Neural Listeriosis	*Listeria monocyotogens* (G +ve coccobacilli)	Drooping ears, circular movements by holding head to one side, protrusion of tongue, Microabscesses in brain	CSF, brain samples	Immunofluorescence, Histopathology	Ampicillin, Amoxycillin, Gentamicin
9	Tuberculosis	*Mycobacterium tuberculosis, Mycobacterium bovis, Mycobacterium avium* (Aerobic Acid fast bacilli)	Loss of condition, Mastitis, Tubercles in the affected lymphnodes,	Lymphnodes, tissue lesions, aspirates, nasal discharges	Acid fast staining, Tuberculin test, Gamma interferonassay	Rifampicin, Isoniazid, streptomycin, Paraamino salicylic acid
10	Paratuberculosis / Johne's Disease	*Mycobacterium avium* ssp *paratuberculosis* (Aerobic Acid fast bacilli)	Persistant & profuse diarrhoea, Progressive weight loss, emaciation	Faeces, rectal scraping / pinch biopsies, serum	Acid fast staining method, Johnin test, CFT, AGID	Isoniazid, rifampin, Streptomycin, Clofazimine, Dapsone
11	Caseous Lymphadenitis (Sheep & Goat) Ulcerative Lymphangitis (Horse & Cattle)	*Corynebacterium pseudotuberculosis* (G +ve pleomorphic bacilli)	Abscess in lymph nodes & lymphatic vessels with discharge of pus	Pus, Abscess, Nasal & oral secretions	Staining & Culture	Penicillin, Tetracycline, Erythromycin
12	Pizzle Rot / Ulcerative balanoposthitis (Sheep)	*Corynebacterium renale* (G +ve pleomorphic bacilli)	Pyelonephritis with turbid or blood stained urine, brownish crusts around the preputial orifice	Urine, Prepucial mucus membrane	Staining & Culture	Penicillin, Tetracycline
13	Bovine Actinomycosis / Lumpy Jaw	*Actinomyces bovis* (G +ve filamentous bacilli)	Swelling with discharge of pus, suppurative & proliferative osteomyelitis	Pus, Abscess	Staining Culture & Radiography	Penicillin, Streptomycin, Tetracyclin, Cephalosporins
14	Colibacillosis / White Scours / Colisepticaemia	*Escherichia coli* (G –ve coccobacilli)	Greyish white diarrhoea with fetid odour	Faeces, Intestinal content, Tissues lesion, Blood	Culture, Biochemical test, Serotyping	Amoxycillin, sulponamide, Tetracycline, Fluoroquinolones
15	Hjarre's Disease / Coligranuloma (Birds)	*Escherichia coli* (G –ve coccobacilli)	Diarrhoea, respiratory distress, granulomatous lesions in intestines	Granulomatous lesions, Blood	Culture, Biochemical test, Serotyping	Amoxycillin, sulponamide, Tetracycline, Fluoroquinolones

Sl. No.	Disease / Condition	Causative agent	Clinical signs	Samples	Diagnostic tests (Acc. to OIE)	Treatment
16	Terminal Dry Gangrene (Calves)	*Salmonella dublin* (G –ve coccobacilli)	Gangrene in affected terminal regions	Tissue lesions, Blood	Culture, Biochemical test, Serotyping	Chloramphenicol, Streptomycin, Ampicillin
17	Bacillary White Diarrhoea / Pullorum Disease (Chicks)	*Salmonella pullorum* (G –ve coccobacilli)	Huddling near heat source, Whitish diarrhoea, Pasty vent	Faeces, Intestinal content, Blood	Agglutination test (Plate or tube test)	Chloramphenicol, Streptomycin, Ampicillin
18	Fowl Typhoid (Birds)	*Salmonella gallinarum* (G –ve coccobacilli)	Diarrhoea with greenish coloured faeces, Paleness of wattle & comb	Faeces, liver, Spleen, Blood	Agglutination test (Plate or tube test)	Chloramphenicol, Streptomycin, Ampicillin
19	Fowl Cholera	*Pasteurella multocida* type A & F (G –ve coccobacilli)	Swollen, cyanotic joints, wattles & comb	Blood, Bone marrow, Liver, Spleen	Staining, AGID, CIE	Penicillin, Tetracycline,
20	Haemorrhagic Septicaemia	*Pasteurella multocida* type B & E (G –ve coccobacilli)	Respiratory distress, Purulent nasal discharge, brisket oedema, sudden death	Blood, Oedema fluid, Nasal discharges, Lung, Liver	Staining (Bipolar organisms), AGID, CIE	Penicillin, Tetracycline
21	Shipping Fever (Cattle & Sheep)	*Mannheimia haemolytica* (G –ve coccobacilli)	Respiratory distress, Bronchopneumonia & pleurisy	Blood, Nasal discharges, Lung, Liver, Spleen	Staining (Bipolar organisms), AGID, CIE	Oxytetracycline, Potentiated Sulphonamides, Ampicillin
22	Kennel Cough	*Bordetella bronchiseptica* (G –ve bacilli)	Sever Coughing, gagging or retching	Tracheal aspirate & exudates, Nasal swab	Slide haemagglutination test	Amoxycillin, Tetracycline, Fluoroquinolones
23	Wooden Tongue / Timber Tongue (Cattle)	*Actinobacillus ligneresii* (G –ve pleomorphic bacilli)	Enlarged, protruded tongue with Pyogranulomatous lesions discharging pus	Pus, Biopsy material, exudates	Clinical signs and culture	Potentiated sulphonamides, Penicillin & streptomycin
24	Swine Pleuropneumonia	*Actinobacillus pleuropneumoniae* (G –ve pleomorphic bacilli)	Blood stained froth around nose & mouth, Respiratory distress, Extensive haemorrhages and fibrin deposit in lungs	Tacheal washing, Nasal discharges, Affected tissues	Staining & Culture	Sulfonamides, Long acting tetracyclines, penicillins

Sl. No.	Disease / Condition	Causative agent	Clinical signs	Samples	Diagnostic tests (Acc. to OIE)	Treatment
25	Infectious Coryza (Poultry)	*Avibacterium paragallinarum* (G –ve coccobacilli)	Oedema of face, Nasal discharge, sneezing,	Oedema, Nasal discharges, Swabs	Immunoperoxidase, AGID	Oxytetracycline, Sulphonamides
26	Contagious equine metritis	*Taylorella equigenitalis* (G –ve coccobacilli)	Copious mucopurulent vaginal discharge after few days of service, Temporary infertility	Genital swabs, Semen, Vulval discharges	Slide agglutination test, FAT	Nitrofurazone
27	Melioidosis / Pseudoglanders	*Burkholderia pseudomallei* (G –ve aerobic bacilli)	Abscess in lungs, liver, spleen, lymphnodes and subcutis	Abscess, Affected tissues, Blood	Indirect HA, CFT, FAT	Tetracycline, Trimethoprim, Amoxicillin – clavulanic acid
28	Glanders (Horse)	*Burkholderia mallei* (G –ve aerobic bacilli)	Respiratory distress with nasal discharges, small cutaneous or sub cutaneous nodules in lymphatic vessels of limbs & flanks (Farcy)	Pus, Nasal discharges, Blood, Lymphnodes	Mallein test, CFT	Tetracycline, Fluoroquinolones Gentamicin, Sulfonamides
29	Brucellosis / Bangs Disease/ Abortion Storm (Cattle & Sheep)	*Brucella abortus, Brucella melitensis* (G –ve coccobacilli)	Abortion in last trimester, retained placentae	Serum, Aborted foetal stomach contents, placenta, Cotyledons	ELISA, MZN staining, AMOS - PCR, CFT, BBAT, FPA	Tetracycline, Rifampin, Fluoroquinolones Aminoglycosides
30	Chronic Respiratory Disease (Birds)	*Mycoplasma gallisepticum* (Pleomorphic bacteria)	Respiratory distress, Pneumonia, air sacculitis	Pleural fluid, lung tissue	HI, FAT, CFT, Plate agglutination test	Tylosin, Tetracycline, Erythromycin,
31	Contagious Bovine Pleuropneumonia (CBPP)	*Mycoplasma mycoides* ssp *mycoides* (Pleomorphic bacteria)	Respiratory distress, Pneumonia, Pleurisy, polyarthritis	Nasal discharges & swabs, Pleural fluid, lung tissue, lymphnodes	Diene`s staining, Microcolonies, FAT, CFT, Passive HA, Dot blot assay	Tetracycline, Gentamycin, Kanamycin, Tylosin

Sl. No.	Disease / Condition	Causative agent	Clinical signs	Samples	Diagnostic tests (Acc. to OIE)	Treatment
32	Contagious Caprine Pleuropneumonia (CCPP)	*Mycoplasma capricolum* ssp *capripneumoniae* (Pleomorphic bacteria)	Respiratory distress, Pneumonia, Pleurisy, polyarthritis	Nasal discharges & swabs, Pleural fluid, lung tissue,	Diene`s staining, Microcolonies, FAT, CFT, Passive HA, Dot blot assay	Tetracycline, Gentamycin, Kanamycin, Tylosin
33	Leptospirosis	*Leptospira interrogans* ssp. (Helical curved bacteria with hooked ends)	Hemorrhages, Jaundice, haemoglobinuria, nephritis, anaemia, abortion, Mastitis	Blood, Urine, Kidney, Foetal kidney smears, Cotyledons, Uterine discharges	Sliver impregnation techinque, Microscopic agglutination test, FAT, DFM	Amoxycillin, Penicillin, Dihydro streptomycin, Tetracycline
34	Lyme Disease (Humans, dog, cattle)	*Borrelia burgdoferi* sensu lato (Helical curved bacteria)	Fever, arthritis, cardiac & neurologic disturbance	Synovial fluid, Serum, Affected tissues smears	Silver impregnation technique, FAT	Amoxycillin, Oxytetracycline
35	Avian Spirochaetosis	*Borrelia anserina* (Helical curved bacteria)	Fever, marked anaemia, weight loss	Buffy coat, Blood or tissue smears	Silver impregnation technique, FAT	Amoxycillin, Oxytetracycline
36	Genital Campylobacteriosis (Cattle & Sheep)	*Campylobacter fetus ssp. Veneralis & ssp. Fetus* (Slender, curved G –ve bacilli)	Temporary infertility, early embryonic death, return to oestrus at irregular periods, Abortion (Ewes)	Faeces, semen, cervical mucus, Prepucial, urethral swabs	Vaginal mucus agglutination test, IFAT	Dihydro streptomycin, Oxytetracycline, Enrofloxacin
37	Q-Fever	*Coxiella burnetti* (Small pleomorphic bacilli)	Influenza like symptoms, Endocarditis (Chronic infection)	Faeces, uterine discharges, Placenta, cotyledons, Milk,	MZN staining, FAT, Cultivated in cell lines & embryonated egg inoculation	Tetracyclines, Chloramphenicol
38	Anaplasmosis (Ruminants)	*Anaplasma bovis, A. marginale, A. central A. ovis* (Pleomorphic G -ve bacteria)	Marked anaemia, jaundice, weight loss	Blood	IFAT, CFT, Giemsa staining	Long acting oxytetracyclines, imidocarb dipropionate
39	Canine Ehrlichiosis	*Ehrlichia canis* (Pleomorphic G -ve bacteria)	Fever, thrombocytopenia, leucopenia, anaemia	Blood	FAT, Giemsa staining	Doxycycline, Oxytetracycline

Sl. No.	Disease / Condition	Causative agent	Clinical signs	Samples	Diagnostic tests (Acc. to OIE)	Treatment
40	Avian Chlamydiosis / Psittacosis / Ornithosis	*Chlamydophila psittaci* (G -ve intracellular bacteria)	Nasal & ocular discharges, diarrhoea, respiratory distress	Respiratory discharges, faeces, affected tissues	CFT, Immunohistochemistry, In-vivo culturing	Tetracyclines
41	Ovine Enzootic Abortion	*Chlamydophila abortus* (G -ve intracellular bacteria)	Abortion, pink uterine exudates, Infertility	Placenta, cotyledons, Uterine discharges	CFT, Indirect immunofluorescence, In-vivo culturing	Long acting oxytetracycline
42	Foot Rot (Sheep & Cattle)	*Dichelobacter nodosus, Fusobacterium necrophorum* (G –ve anaerobic bacteria)	Lameness, Necrosis of the foot	Foot discharges or lesions	Culture	Foot soaks, Long acting penicillin, Oxytetracycline, Streptomycin
43	Mastitis	*Staphylococcus spp. Streptococcus spp, E. coli, Klebsiella, Proteus, Corynebacteria spp, Brucella spp,*	Inflammation of udder, reduction in milk yield	Milk	Demonstration of organisms in milk smears	Based on ABST (Enrofloxacin, Ciprofloxacin, Gentamycin)

2. FUNGAL DISEASES

Sl. No.	Disease / Condition	Causative agent	Clinical signs	Samples	Diagnostic tests	Treatment
1	Dermatophytosis (Ringworm infection)	*Microsporum spp.* *Trichophyton spp.* *Trichosporon spp.* *Epidermophyton spp.*	Alopecia, ringworm lesions, scaling, broken hairs	Hair, skin & claw scrapings, biopsy material	Wood`s lamp test, Macroconidia (Lactophenol stain), Hair perforation test	Itraconazole, Fluconazole, Terbinafine, Griseofulvin
2	Crop mycosis / Thrush / Avian moniliasis (Birds)	Candida albicans (Yeast)	Thickened crop wall with yellowish grey necrotic corrugated pseudomembrane (terry-towelling appearance)	Biopsy, PM tissue samples	Germ or serum tube test	Nystatin, Amphotericin B, Ketoconazole, clotrimazole
3	Brooder's pneumonia (Chicks)	*Aspergillus fumigatus*	Yellowish nodules on lungs & air sacs	Lung tissues	Shape of spore heads of aspergillus, Lactophenol staining	Itraconazole, Fluconazole
4	Mycotic abortion & mastitis (Cattle)	*Aspergillus fumigatus*	Aborted foetus with raised cutaneous plaques, intercotyledonary areas necrotic, thickened & leathery	Milk, uterine discharges, Foetal abomasal contents	Shape of spore heads of aspergillus, Lactophenol staining	Itraconazole, Fluconazole
5	Guttural pouch mycosis (Horses)	*Aspergillus fumigatus*	Epistaxis, dysphagia, laryngeal hemiplegia, postauricular swelling & unilateral nasal discharge	Respiratory discharges	Radiography, Histopathology, Shape of spore heads of aspergillus	Itraconazole, Fluconazole
6	Cryptococcosis	*Cryptococcus neoformans spp.*	Nasal discharges, Neurological signs	Exudates, CSF, biopsy or PM tissues	Capsule staining, growth of bird seed agar, Urease test	Amphotericin B, fluconazole, Itraconazole,
7	Blastomycosis	*Blastomyces dermatitidis*	Coughing, exercise intolerance, dyspnoea	Respiratory discharges	CIE, Histopathology, FAT, Giemsa /MB staining	Itraconazole, Amphotericin B
8	Histoplasmosis	*Histoplasma capsulatum*	Chronic coughing, persistent diarrhoea, emaciation	Respiratory discharges	FAT, Histopathology, Histoplasmin skin test	Itraconazole, ketoconazole, Amphotericin B
9	Mycotoxicoses	A. *flavus,* A. *parasiticus, A. ochraceus, Claviceps purpura, Fusarium spp.*	Sudden death	Feed & tissues	HPLC, UPLC, TLC, Mass spectrometry	Limit the contamination of food with fungi

3. VIRAL DISEASES

Sl. No	Disease / Condition	Causative agent	Clinical signs	Sample collected	Diagnostic tests (Acc. to OIE)
1	Pox disease (Cattle, Buffalo, Sheep, Goat, Fowl, Camel, mouse)	Poxvirus (Respective species virus) (Poxviridae, Enveloped ds DNA virus)	Raised pock lesions anywhere on the body surfaces	Lesion scrapings and dried scabs	AGID, FAT, VN
2	Lumpy skin disease (Pseudourticaria) (Cattle)	Capri poxvirus (Poxviridae Enveloped ds DNA virus)	Nodular lesions in the skin with necrosis, lymphadenitis and edema of the limbs	Lesion scrapings and dried scabs	VN, IFAT
3	Contagious pustular dermatitis (Orf / Scabby mouth) (Lambs and kids)	Parapox virus (Poxviridae, Enveloped ds DNA virus)	Pox lesions on the lips – proliferative type of lesions on lips & mouth	Skin lesions and scab	VN, AGID, FAT
4	Infectious bovine rhinotracheitis (Cattle)	Bovine herpes virus 1 (Herpesviridae, Enveloped ds DNA virus)	Respiratory symptoms, abortions, vulvo vaginitis in cows and balanoposthitis in bulls	Nasal, tracheal & vulval swabs, Aborted fetal liver, spleen and stomach contents paired sera	VNT, FAT
5	Infectious laryngo tracheitis (chicken)	Gallid herpesvirus 1 of Iltovirus (Herpesviridae, Enveloped ds DNA virus)	Respiratory distress, expectoration of blood stained exudates, extended neck (Pump handle respiration), lacrimation and swelling of infra orbital sinus	Paired sera, tracheal exudates, larynx, trachea, lungs	AGID, VNT, IFT
6	Pseudo rabies (Mad itch / Aujesky disease) (Pigs, cattle, dogs)	Porcine Herpesvirus 1 (Herpesviridae, Enveloped ds DNA virus)	Coughing, pneumonia, paralysis and convulsions. In cattle – intense pruritis	Lung, tonsils, kidney, spleen or brain of pigs, Nasopharyngeal swabs, lungs of affected cattle.	VNT
7	Marek's disease (chicken)	Gallid herpesvirus 2 of Mardivirus (Herpesviridae, Enveloped ds DNA virus)	Athlete's posture, Torticolis, paralysis, Feather follicular lesions.	Paired sera, feather tips & follicle, blood, visceral organs having lymphomas, a portion of brachial plexus / sciatic nerve	AGID
8	Malignant catarrhal fever (Catlle, buffalo, deer)	Bovine herpesvirus 3 (Herpesviridae, , Enveloped ds DNA virus)	Degenerative changes in the URT and alimentary tract, diffuse necrosis and ulcers in oral and nasal mucosa, enlargement of lymphnodes	Blood, Lymphnode, thymus, spleen, bone marrow, ocular and nasal secretions.	VN, IFAT, Immunopero xidase test,

Sl. No	Disease / Condition	Causative agent	Clinical signs	Sample collected	Diagnostic tests (Acc. to OIE)
9	Equine rhino pneumonitis (Equine abortion)	Equine herpesvirus 1 (Herpesviridae, Enveloped ds DNA virus)	Catarrhal respiratory, Aborted fetus with petechial of mucous membranes, excessive pleural fluid, spleenic enlargement and necrosis of liver	Paired sera, Deep nasal swabs, vaginal swabs, lungs, liver, spleen of aborted fetus	VNT
10	Infectious CanineHepatitis (Dogs & Foxes)	Canine adenovirus 1 (Adenoviridae, Non-enveloped ds DNA virus)	Vomition, Mucopurulent oculonasal discharge, Corneal opacity (Blue eye). Acute encephalitis In foxes	Blood, Paired sera, urine, liver, spleen and lymphnodes	FAT, CFT
11	Egg drop syndrome –76 (EDS-76) (Chicken)	At adenovirus (Adenoviridae, Non-enveloped ds DNA virus)	Sudden drop in egg production. Change in quality of eggs - thin shelled, shell less, misshapen	Paired serum, Cloacal swabs, ovary, oviduct 3	HI, AGPT
12	Papillomas (Warts)	Papillomavirus (Papillomaviridae, Non-enveloped ds DNA virus)	Thickened rough nodules, sometimes large peduncle cauliflower like growth usually on head, neck, shoulder, udder	Biopsy or the lesions	Lesions, Chick embryo inoculation
13	Canine Parvo viral diarrhoea	Canine Parvovirus 2 (Parvoviridae, Non-enveloped ss DNA virus)	Vomiting, Haemorrhagic foul smelling diarrhoea, myocarditis.	Fresh faecal samples, intestinal scrapings, Paired serum	HA, HI, FAT, LFA
14	Chicken infectious anemia	Circovirus (Circoviridae, Non-enveloped ss DNA virus)	Anaemia, aplasia of bone marrow, lymphoid depletion, atrophy of thymus and bursa, enlarged liver.	Buffy coat cells, whole blood for PCV, thymus, bone marrow, liver and spleen	Blood picture, FAT
15	Blue tongue (Sheep)	Orbivirus (Reoviridae, Non-enveloped segmented ds RNA virus)	Oculonasal discharge, hyperemic mouth mucosa, dental pad and lips. Tongue swollen, cyanotic, ulcerative, lesions on coronary band	Blood, lesion scrapings, nasal & faecal swabs, spleen and Lymphnodes from dead animals paired sera	IFT, VNT
16	Infectious Bursal Disease (Chicken, turkey, ducks)	Avibirnavirus (Birnaviridae, Non-enveloped ds RNA virus)	Severe depression, ruffled feathers, trembling, pecking the vent region, diarrhea	Paired serum, affected bursa, spleen, kidney and liver	AGID, FAT, VNT
17	Bovine viral diarrhoea and Mucosal disease (Cattle)	Pestivirus (Flaviviridae, Enveloped ss +ve sense RNA virus)	Mucopurulent nasal discharge, greyish white ulcers in the mouth, foul smelling diarrhoea.	Blood, paired sera, faecal samples, nasal exudates, mesenteric lymphnode	VNT, Immunohisto chemistry
18	Classical Swine Fever (Hog Cholera)	Pestivirus (Flaviviridae, Enveloped ss +ve sense RNA virus)	Leukopenia, conjunctivitis, dyspnoea, vomition, diarrhoea, discoloration of the abdomen, ears and snout. In PM button like ulcers in Large intestine	Blood, paired sera, nasal & eye swabs, Mesenteric lymphnodes, spleen and tonsil from dead animals	IFAT, VNT

Sl. No	Disease / Condition	Causative agent	Clinical signs	Sample collected	Diagnostic tests (Acc. to OIE)
19	Avian infectious bronchitis	Corona virus (Coronaviridae, Enveloped ss +ve sense RNA virus)	Coughing, gasping in birds of all ages, facial swelling, severe drop in egg production, deterioration in the quality of eggs like small size, misshape	Paired serum lungs and tracheal scrapings & swabs, caecal tonsils, kidney	VNT, HI
20	Foot and Mouth Disease/ Apthous fever (All cloven footed animals)	Aptho virus (Picornaviridae, Non-enveloped ss +ve sense RNA virus)	Stringy salivation, smacking of lips, vesicles of the tongue, gums, dental pads, cheeks, teats, around the muzzle, coronary band and interdigital cleft	Vesicular fluid, Epithelium from lesions of the mouth, Blood, serum, Lymphnodes and heart from dead animals	VNT
21	Avian leucosis (Lymphoid leukosis / Big liver disease) (chicken)	Avian leucosis virus (Retroviridae, Enveloped ss +ve sense RNA virus)	Develop only in chicks above 16-20 weeks of age, nodular lesions in visceral organs	Paired sera, affected portion	COFAL, Indirect FAT
22	Visna/Maedi (Sheep & Goats)	Lentivirus (Retroviridae, Enveloped ss +ve sense RNA virus)	Maedi: Respiratory distress. Visna: Lameness, slight tremor of head and occasional blindness.	Lungs, Spleen, sera, Lymphnode, Salivary glands, buffy coat cells, mammary glands	AGID
23	Equine infectious anaemia (Horse, mule & ass)	Lentivirus (Retroviridae, Enveloped ss +ve sense RNA virus)	Anaemia, petechial haemorrhage in visible mucous membrane, enlargement of lymphnodes	Blood, paired serum samples	AGID (Coggin's test)
24	Peste des Petits Ruminants (PPR) (Sheep and Goat)	Morbilivirus (Paramyxoviridae, Enveloped ss -ve sense RNA virus)	Nasal catarrhal discharges, Mucosal erosion, intermittent diarrhoea	Blood, Serum, Nasal discharges, intestinal mucous membrane, lungs, tonsils & LNs from dead	AGID, CIE, VNT, HI
25	Canine distemper	Morbilivirus (Paramyxoviridae, Enveloped ss -ve sense RNA virus)	Bronchopneumonia, vomiting, diarrhoea, vesicular and pustular dermatitis	Blood, paired sera Nasal or conjunctival secretions/ epithelium, corneal impression smears and swabs	VNT, FAT, CFT
26	New castle disease (Ranikhet disease)	Avulavirus (Paramyxoviridae, Enveloped ss -ve sense RNA virus)	Respiratory distress, Muscle tremors, torticolis, paralysis of legs and wings, reduced egg production	Paired sera, spleen, brain, lungs, liver, trachea, proventriculus	HI
27	Avian influenza (Fowl pest / Fowl plague)	Orthomyxovirus (Orthomyxoviridae, Enveloped segmented ss –ve sense RNA virus)	Cessation of egg laying, respiratory distress, lacrimation, sinusitis, diarrhea, oedema of the head, face and neck, cyanosis of comb and wattles, torticolis,	Blood, Paired sera, Nasal Swabs, lungs, liver, spleen	VNT, HI

Sl. No	Disease / Condition	Causative agent	Clinical signs	Sample collected	Diagnostic tests (Acc. to OIE)
28	Rabies (All warm blooded animals)	Lyssavirus (Rhabdoviridae, Enveloped ss -ve sense RNA virus)	Rapidly fatal encephalitis, Excitement followed by paralysis; death in 3-6 days following the onset of clinical symptoms	Impression smears & portions of hippocampus, cerebellum and ammons horn fixed in absolute ethanol or 10% acetone or a portion of the brain	FAVN, FAT, dRIT, RFFIT
29	Ephemeral fever (Cattle)	Bovine ephemeral fever virus (Rhabdoviridae, Enveloped ss -ve sense RNA virus)	Muscular shivering, lameness, stiffness of joints and enlargement of peripheral lymphnodes	Blood, Lymphnodes	VNT
30	Viral enteritis in young animals	Rotavirus, Coronavirus, Astrovirus	Diarrhoea in young animals, faeces may be watery, brown, grey or light green; sometimes blood and mucus	Faeces, intestinal contents	FAT, LFA
31	Scrapie / Bovine spongiform encephalopathy	Prions	Tremors, hyperesthesia, abnormal posture, hind limb ataxia, reduced milk yield, and weight loss.	Brain tissue	Histopatholgy Immunohisto chemistry, Western blotting

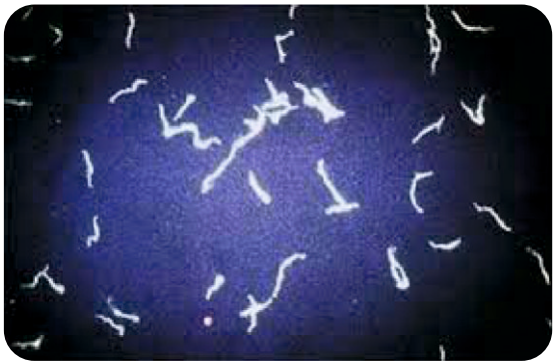
Note: 1. Isolation of organisms is gold standard for diagnosis of all disease conditions
2. ELISA & PCR are the commonly used tests for diagnosis of all organisms
3. Treatment for all viral diseases is symptomatic

(ABST: Antibiotic sensitivity test, AGID: Agar gel immunodiffusion, BBAT: Buffered Brucella antigen test, CAM: Chorio allantoic membrane, CFT: Complement fixation test, CIE: Counter Immune Electrophoresis, COFAL: Complement fixation test for avian leukosis, DFM : Dark field microscopy, dRIT: Direct rapid immunohistochemistry, ds: double stranded, ELISA: Enzyme linked immunosorbent assay, FAVN: Fluorescent antibody viral neutralization, FAT: Fluorescent antibody technique, FPA: Fluorescence polarisation assay, HA: Haemagglutination, HI: Haemagglutination inhibition, IPT: Immunoperoxidase test, LFA : Lateral flow assay, MZN: Modified Ziehl Neelsen, PAGE: Polyacrylamide gel electrophoresis, PCR: Polymerase chain reaction, PHA: Passive Haemagglutination, RFFIT: Rapid fluorescence focus inhibition test, RT-PCR: Reverse transcription polymerase chain reaction, ss: single stranded, URT: Upper respiratorytract; VNT: Virus neutralization test)

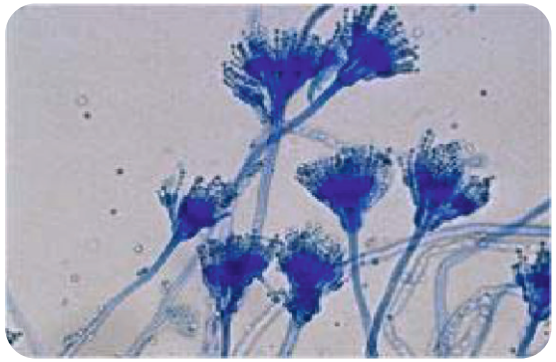
DRUG	DOSAGE
Amoxicillin – clavulanic acid	Dog Cat 8.75 mg/kg BW Tid – IV IM SC 12.5-25 mg/kg BW Bid - Oral
Amoxycillin	Dog Cat 10-20 mg Bid Cattle Horse Goat Pig 10 mg/kg BW Bid
Amphotericin B	Systemic mycoses - Dog 0.25 -0.5 mg/kg BW (every 48 hrs) Histoplasmosis -1.8 mg/kg BW daily for7 days
Ampicillin	Cattle Sheep Goat Horse 5-10 mg/kg BW Qid Dog Cat – 20 mg/kg BW Qid - Oral 10 mg/kg BW Qid - IM IV SC
Cephalosporins	Cefadroxil - Oral, Dog Cat 10 - 30 mg/kg BW Tid Cefalexin - Oral, Dog Cat 10-30 mg/kg BW Bid Cattle Buffalo - 5-10mg/kg BW Bid Cefotaxime - IM IV Dog Cat 25 -50 mg/kg BW/day as divided doses
Chloramphenicol	Bovine - 2-4mg/kg BW Bid IV 20-30- mg/kg BW IM Sheep Goat -10 mg/kg BW Bid IV 30-40 mg/kg BW IM Dog - 25-50 mg/kg BW Bid IV
Chlortetracycline	Large animals 10-20 mg/kg BW IM Small animals 25-50 mg/kg BW IM
Ciprofloxacin	Oral IM IV Dog Cat 5 -15mg/kg BW Cattle sheep 4 - 5 mg/kg BW
Clofazimine	Dog Cat 2-12 mg/kg BW Oral OD for 2 – 6 M
Dapsone	Oral 1 mg/kg BW OD or Tid
Doxycycline	Oral Dog Cat - 5-10 mg/kg bw/day Cattle Sheep Goat 1-2 mg/kg BW
Enrofloxacin	Oral IM IV 2.5-5 mg/kg BW
Erythromycin	Cattle Sheep Goat Horse 2.2-4.4 mg/kg BW Pig 2.2-6 mg/kg BW Dog Cat 10-2- mg/kg BW Bid - Tid
Fluconazole	Dog 2.5 – 5mg mg/kg BW Bid Oral Cat 50 mg / cat Oral
Gentamicin	IM IV Cattle Sheep Goat Pig Dog Cat- 4 mg/kg BW Bid, Poultry - 3-5 mg/kg BW

DRUG	DOSAGE
Griseofulvin	Cattle 7.5-10mg/kg BW/day for 30 days Dog Cat 20mg/kg BW/day for 30 days
Itraconazole	Oral Dog Cat - 5 mg/kg Bw OD upto 4-20 weeks
Kanamycin	SC IM IV Cattle 5-10 mg/kg BW IM IV Dog Cat 5 mg/kg BW IM IV
Ketoconazole	Oral Dog Cat - 10mg/kg BW
Long acting oxytetracyclines	IM IV Cattle Sheep Goat Swine Horse 5- 10mg/kg BW/day, Dog Cat 10-20 mg/kg BW/day
Long acting penicillin	Benzathine penicillin - IM Cattle Horse Sheep Goat Swine - 12000 IU/kg BW, Dog Cat 40000 IU/kg BW
Nitrofurazone	Topical application cream or powder
Norfloxacin	Oral Dog Cat - 22 mg/kg BW Bid
Oxytetracycline	IM IV, Cattle Sheep Goat Pig Horse - 5-10 mg/kg/BW/day Dog Cat 10-20 mg/kg/BW/day
Penicillin	IM IV 10,000 – 20,000 IU/kg BW
Procaine Penicillin	IM Cattle Horse Sheep Goat Pig 4000 – 10000 IU / kg BW Dog 10,000 – 20,000 IU/kg BW
Rifampicin	Oral Dog Cat 10 - 15 mg/kg BW OD
Streptomycin	IM Cattle Horse Goat Dog Cat -10 mg/kg BW
Sulphonamides	All animals - Oral IV IM 50 - 300 mg/kg BW
Terbinafine	Oral Dog Cat 5-50mg/kg BW OD
Tetracycline	Cattle 5-10 mg/kg BW Tid Dog 10-20 mg/kg BW Tid
Tylosin	Oral Cattle Sheep Goat 10-20 mg/kg BW Pig 5-20 mg/kg BW Dog Cat 20-40 mg/kg BW

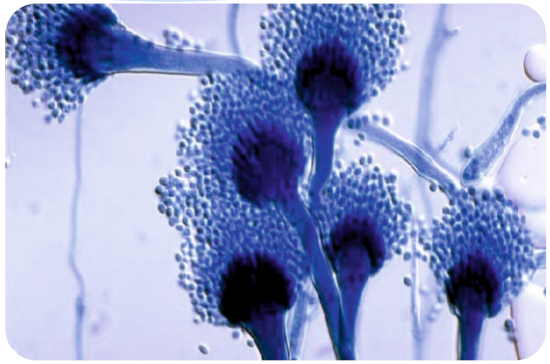
IV: Intravenous, IM: Intramuscular, SC: Subcutaneous, IU; International unit, BW: Body weight, OD: Oral dose, mg: Milligram, Kg: Kilogram, Bid = Twice a day, Tid = Three times a day, Qid = Four times a day



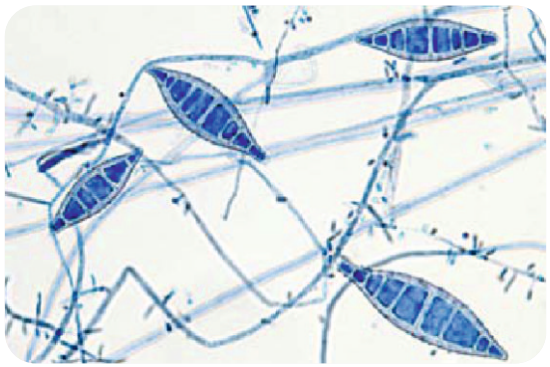
Leptospira
(Darkfield microscopy)



Penicillium
(Lactophenol cotton blue staining)



Aspergillus ssp.
(Lactophenol cotton blue staining)



Microsporum canis
(Lactophenol cotton blue staining)



Microsporum gypseum
(Lactophenol cotton blue staining)



Microsporum nanum
(Lactophenol cotton blue staining)



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